

COMPUTER TECHNOLOGY

Computer technology is an expanding branch of engineering, with roots in both electrical engineering and computer science. It includes the related areas of computer hardware and software; computer interfacing, programming, and networking; analog and digital electronics; and robotics. Computer technicians, technologists, and engineers work in every sector of society, in careers ranging from building and repairing computer systems to designing and installing computer networks, to designing and building prototype robots and electronic devices.

Students taking computer technology courses will learn to work safely with computer equipment, electronic circuits, and robotic devices, and will explore the Essential Skills and work habits that are important for success in computer technology. Computer technology courses prepare students for apprenticeship, further study at college or university, or entry into the workplace directly after graduation.

The list of approved emphasis areas for computer technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Computer Engineering Technology, Grade 12

University/College Preparation

TEJ4M

This course extends students' understanding of computer systems and computer interfacing with external devices. Students will assemble computer systems by installing and configuring appropriate hardware and software, and will learn more about fundamental concepts of electronics, robotics, programming, and networks. Students will examine related environmental and societal issues, and will explore postsecondary pathways leading to careers in computer technology.

Prerequisite: Computer Engineering Technology, Grade 11,
University/College Preparation

A. COMPUTER TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of internal buses and storage devices, and of advances in computer technology;
- A2.** demonstrate an understanding of system optimization and of permissions, attributes, firmware, and communication standards used in computer systems;
- A3.** demonstrate an understanding of devices and electronic circuits in interfaces and control systems;
- A4.** demonstrate an understanding of network addressing and routing;
- A5.** demonstrate an understanding of computer logic circuits and the representation, manipulation, and transmission of data by computers.

SPECIFIC EXPECTATIONS

A1. Computer Hardware

By the end of this course, students will:

- A1.1** describe the function of internal buses within computer systems (*e.g., data bus; memory bus; address bus; buses in CPUs, RAM, and chipsets*);
- A1.2** identify appropriate storage devices for various computing requirements (*e.g., optical drives, flash drives, single and arrayed hard drives*);
- A1.3** describe how advances in computer technology (*e.g., CPUs, memory, storage*) have spurred advances in related technologies (*e.g., cellular telephones, hand-held devices, image capture systems*).

A2. Computer Systems

By the end of this course, students will:

- A2.1** describe how to assign permissions and attributes to drives, folders, and files with various operating systems (*e.g., user permissions, archiving, encryption, compression*);
- A2.2** describe methods for optimizing a computer system (*e.g., updating firmware, updating drivers, defragmenting files, allocating virtual memory*);
- A2.3** describe the functions of the BIOS and other firmware in computer systems (*e.g., boot process, hardware recognition, resource allocation, port settings, energy management*);

- A2.4** describe various standards for data flow between computer ports and peripherals (*e.g., MIDI, RS-232, USB, IEEE 1394, SCSI*).

A3. Electronics, Robotics, and Computer Interfacing

By the end of this course, students will:

- A3.1** use technical terminology to accurately describe the specifications for electronic components and computer interfaces;
- A3.2** describe the function and operation of various input devices, output devices, and electronic circuits used in interface and control systems (*e.g., input devices: temperature sensor, light sensor, position encoder; output devices: AC motor, stepper motor; circuits: power supply, motor driver*);
- A3.3** calculate the values and operating parameters of electronic components in a circuit, using fundamental laws and circuit-analysis techniques (*e.g., Ohm's law, Kirchhoff's laws, Thévenin and Norton equivalent circuits*);
- A3.4** draw and interpret diagrams that use standard symbols to represent electronic components and the operation of control systems (*e.g., schematic diagram, block diagram, flow chart*);
- A3.5** research and select components based on circuit requirements (*e.g., use Internet searches, manufacturer's data sheets, supplier catalogues, and/or parts database*).

A4. Networking Concepts

By the end of this course, students will:

- A4.1** describe the function of routed protocols (*e.g., IP, IPX*) and routing protocols (*e.g., RIP, OSPF, EIGRP*) in the transmission of data over a network;
- A4.2** explain the seven layers of the OSI (open systems interconnection) model and the corresponding network devices;
- A4.3** describe IP addressing and subnetting strategies for IP networks (*e.g., borrowing bits, calculating number of subnets and hosts, determining specific subnet address range*);
- A4.4** describe static and dynamic classful public and private addressing and related strategies (*e.g., Class A, Class B, Class C, NAT, PAT, DHCP*).

A5. Data Representation and Digital Logic

By the end of this course, students will:

- A5.1** perform arithmetic operations on positive and negative binary numbers (*e.g., addition, subtraction*) using two's complement representation;
- A5.2** use Boolean logic (*e.g., Karnaugh maps*) to design a solution to a logic problem that has multiple inputs and outputs (*e.g., manufacturing process, starting a car*);
- A5.3** use Boolean logic and the laws of Boolean algebra to design, simplify, and build computer logic circuits using logic gates (*e.g., adder circuit, decoder circuit*);
- A5.4** describe the role of flip-flop circuits in the storage and flow of data (*e.g., asynchronous counter, synchronous counter, shift register, memory register*);
- A5.5** describe how computers store and work with different types of data, including numbers, characters, and arrays;
- A5.6** explain how analogue quantities can be represented by digital systems (*e.g., analogue-to-digital converter, pulse-width modulation*).

B. COMPUTER TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** build computer systems and connection media to meet specific requirements, using appropriate procedures, tools, and equipment;
- B2.** maintain and troubleshoot a variety of computer hardware and software;
- B3.** design, build, test, and troubleshoot interfaces and other circuits that meet specific design requirements;
- B4.** design, build, configure, maintain, and troubleshoot networks, and set up various network services for users;
- B5.** demonstrate an understanding of programming concepts, and create programs that interact with external devices.

SPECIFIC EXPECTATIONS

B1. Hardware Solutions

By the end of this course, students will:

- B1.1** select appropriate components and build computer systems that meet specific requirements (*e.g., gaming system, engineering workstation, media centre, control system, home office system*);
- B1.2** select and use appropriate procedures, tools, and diagnostic equipment when assembling computing devices (*e.g., procedures: use of anti-static wrist strap and/or mat; tools: crimping; diagnostic equipment: multimeter, cable tester, oscilloscope*);
- B1.3** construct and test connection media for interfacing a computer with an external device (*e.g., serial cable, parallel cable*).

B2. Computer Systems

By the end of this course, students will:

- B2.1** use a variety of sources to collect information for solving computer problems (*e.g., Internet searches, technical reference materials*);
- B2.2** create a log and/or engineering journal to document work done on computer systems (*e.g., troubleshooting, software updates, hardware installations, maintenance*);

- B2.3** use a logical approach to diagnose and troubleshoot computer hardware and software problems.

B3. Electronics, Robotics, and Computer Interfacing

By the end of this course, students will:

- B3.1** use a design process (see pp. 22–23) and appropriate software (*e.g., circuit simulation software, CAD [computer-aided design] software*) to design circuits;
- B3.2** construct circuits made from both discrete components and integrated circuits to perform specific functions (*e.g., regulated power supply, electronic dice, audio amplifier, microcontroller-based alarm circuit*);
- B3.3** safely construct electronic circuits for interfacing or robotic applications using appropriate materials, tools, and techniques, including soldering (*e.g., materials: breadboard, printed circuit board, etchant, solder; tools: soldering iron, etch bath, third hand with magnifier*);
- B3.4** test and troubleshoot electronic circuits, using appropriate methods (*e.g., isolating and substituting components*) and test equipment (*e.g., multimeter, oscilloscope, logic probe*), and modify the circuits to meet design requirements if necessary.

B4. Network Setup and Management

By the end of this course, students will:

- B4.1** design and build a network (*e.g., connection media, interconnection devices, peripherals, server, workstations*) that meets user requirements;
- B4.2** optimize and maintain a computer network (*e.g., check performance, accessibility, and security*);
- B4.3** implement various network services for users (*e.g., HTTP, FTP, remote desktop, SMTP, DHCP*);
- B4.4** configure workstations, servers, and/or networked devices (*e.g., create users, assign privileges to folders, set up services, format and partition hard drives*);
- B4.5** apply logical troubleshooting techniques, using data from simulation and/or diagnostic tools (*e.g., simulation software, packet sniffers, cable tester*).

B5. Computer Programming

By the end of this course, students will:

- B5.1** compare low-level and high-level programming languages;
- B5.2** apply programming concepts including subroutines, parameter passing, decision and repetition structures, arrays, and character representation;
- B5.3** use a design process (see pp. 22–23) to create a program that interacts with a real-world device (*e.g., traffic light, alarm system, robot, joystick*);
- B5.4** write a low-level program that runs on a real or simulated controller device (*e.g., programmable logic controller [PLC], microcontroller, assembler simulator*).

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** analyse environmental issues related to the widespread use of computers and associated technologies, and apply strategies to reduce environmental harm from computer use;
- C2.** analyse societal issues related to the widespread use of computers and associated technologies.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** assess the effects of computer and electronics technology on the environment (*e.g., hazardous materials contained in computer components, use of energy and other resources, fuel consumption and air pollution reduced by computerized traffic-control systems*);
- C1.2** outline and apply strategies to recycle or reuse computers and computer components (*e.g., develop a local recycle/reuse program, create an in-school public awareness campaign*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** assess the benefits of computer and electronic technology for society (*e.g., improved access to technology for economically disadvantaged people and nations; greater efficiency and lower costs for information services; development of a “global village”; software that can help monitor or predict changes in wetland area, deforestation, and climate*);
- C2.2** assess the drawbacks of computer and electronics technology for society (*e.g., Internet gambling addictions, more sedentary lifestyle, spam, telemarketing, loss of privacy, infringement of intellectual property rights through unlicensed copying and electronic distribution*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** explain the importance of safety standards and practices, and use appropriate techniques to avoid health and safety problems;
- D2.** describe ethical and security issues related to the use of computers and related technology;
- D3.** assess career opportunities related to computer technology and electronics, and explain the importance of postsecondary education and lifelong learning in the computer technology industry.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** explain the importance of following industry health and safety standards and practices (*e.g., standards and regulations specified in the Workplace Hazardous Materials Information System [WHMIS] and the Electrical Safety Code; practices such as electrical grounding and precautionary measures when working with live circuits and devices that store electrical energy; ergonomically sound workplace arrangements and practices*);
- D1.2** evaluate and use appropriate techniques to avoid health and safety problems (*e.g., repetitive strain injuries, eye strain, electrical shock, burns from soldering tools*) when assembling, using, and maintaining computer systems.

D2. Ethics and Security

By the end of this course, students will:

- D2.1** describe the components of an acceptable-use policy for computers, cellular technology, PDAs, and/or other electronic devices (*e.g., appropriate use, protection of intellectual property rights, prohibition of plagiarism*);
- D2.2** outline a purchasing policy for computers, taking ethical issues into account (*e.g., the environment, human rights, child labour*);
- D2.3** describe methods to safeguard confidential data and other sensitive electronic information (*e.g., password protection, encryption, biometrics, behaviorometrics, steganography*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** assess various career opportunities related to computer technology and electronics (*e.g., computer engineering technician or technologist, electrical engineer, programmer, systems analyst*), and identify opportunities for further training and certification (*e.g., college or university programs, trade certifications*);
- D3.2** explain the need for lifelong learning in the computer technology industry (*e.g., rapid changes in technology, employability, progress into positions of greater responsibility*);
- D3.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (OSP) (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D3.4** demonstrate an understanding of and apply the work habits that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy*);
- D3.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in computer technology (*e.g., Passport to Safety certificate, OSP Work Plan, OSP Transition Plan, circuit diagrams, photographs of projects, video of working robot*), and explain why having a current portfolio is important for career development and advancement.